

Velocity Vortex
Enterprise Folio
National Final 2026

velocity vortex



Re-Engineering Australia
Foundation



John Gardner MP
MEMBER FOR MORIALTA



BocaBearings

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PALADIN





Initiation Processes

Team Formation
The team formation was a **structured** and **deliberate** process where all members formally committed to progressing to the National final, ensuring a shared vision and alignment of goals from the outset.

Kick Off-Meeting
On establishment of the team and assignment of roles, an official kick-off meeting was held to launch the project, outline key objectives & timelines.

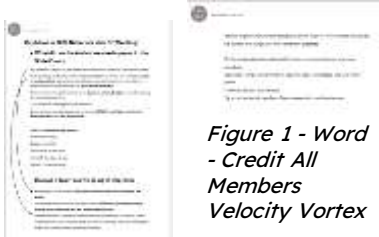
Reflection
Reflecting on past competition experience enabled us to create a targeted improvement plan.

Kick Off Meeting

Following our State Final success, a kick-off meeting on 8 December 2025 launched the project, applying lessons learned and setting direction for the season. Roles were confirmed, goals defined, and key documents—including the Scope Statement and Project Charter—were developed.

Reflection and Areas for improvement:
While we performed strongly in several areas, we identified improvements in workload distribution, communication consistency, and meeting deadlines.

We redistributed workloads, introduced regular meetings, and used Gantt charts and Google Drive to improve timeline management, accountability, and collaboration.



Project Scope Statement

Developed during our kick-off meeting alongside the Project Charter, the Scope Statement defined the project’s focus, key goals, assumptions, and deliverables, while establishing acceptance criteria for each output.

Velocity Vortex 2026 STEM Racing National Scope	
Project Scope	In Scope: •Deliver all required project deliverables to a competition ready standard •Maximise scoring potential across all assessed areas •Win STEM Racing Australian National Final Out of Scope: •Enter 2026 STEM Racing World Final and assist future MHS teams.
Project Objectives	•Develop a new, more refined Trade Display Design •Finalise all folios prior to submission deadlines •Strengthen and unify team identity and branding •Implement consistent uniforms and merchandise •Secure sponsorships and sufficient funding for success of the project •Manufacture two reliable cars achieving the 50.5g mass target
Project Deliverable List	•Four competition portfolios: Enterprise, Engineering, Compliance, and Trade Display Development Brief. •Two competition-ready race cars. •Four branded uniforms with sponsor integration •A professional trade display showcasing team development, branding, and merchandise.
Project Assumptions	•All team members will fulfill their designated roles to a high standard. •All team members will adhere to competition regulations and code of conduct. •Full team availability for the National Final.

Velocity Vortex Project Charter 2026									
Project Title	Velocity Vortex Stem Racing National Final				Project Constraints				
Project Manager	Vivaan Rajpal	Due Dates	Competition: 20-22/03/26						
Project Teacher	Karl Read		Folio Submissions: 13/03/26						
Project Descriptions	The STEM Racing project requires the design, manufacture, testing, and racing of a regulation-compliant miniature F1 car using CAD software. Operating within technical, budget and timeline constraints, the team delivers competitive engineering performance along with professional portfolios, trade display, and presentation outcomes, while developing real-world project management, technical and professional skills.					Acceptance Criteria			
					Clear acceptance criteria were implemented to ensure all deliverables met competition regulations and internal standards. Each output underwent structured review and sign-off before final submission. Refer table below.				
Project Objectives	In Scope: - Design, manufacture, and test a compliant race car - Develop Engineering, Enterprise and Compliance folios - Trade Display, branding, presentation, preparation and Design Brief - Financial management and stakeholder engagement Out of Scope: - Participation in 2026 STEM Racing World Final - Activities outside of the National competition framework and regulations				Project Stakeholders				
					Internal Stakeholders: - Team Members - School Leadership and Mentors External Stakeholders: - Sponsors - Collaborators - Industry Mentors - Competition Organisers & Judges				
					Project Budget				
					Cars	\$400	Accommodation	\$1500	
					Uniforms	\$250	Entrance Fee	\$1600	
					Pit Display	\$1200	Total	\$4950	
					Project Approval	Project Manager Sign : 			Project Teacher Sign: 

Team Members	Team Leader: Vivaan Rajpal Lead Engineer: Liam Bairstow Graphics Designer: Sebastian Richards
Project Milestones	Trade Display Manager: Riley Hutchinson • Start Date & Kick Off Meeting • First sponsor acquired • First car design finalised and machined • MHS Open Night (Marketing) • Trade Display design finalised and materials ordered • Uniforms finalised and ordered • All cars fully manufactured/assembled • Australian National Final
Project Deliverables	• 2x Race Cars • 4 Uniforms • Trade Display • Marketing Materials • Engineering, Enterprise & Compliance Folios • Website
Project Justification	We chose to participate in STEM Racing to experience a real-world engineering and enterprise challenge. Beyond racing, the program develops technical innovation, project management, financial literacy and professional communication, strengthening our industry readiness and future career pathways.

Velocity Vortex Deliverable Acceptance Criteria	
Enterprise Folio	10 x A3 pages of content that meets judging criteria and is presented with clear, professional formatting.
Competition Cars	2 x Race Cars, 1 x Display Car must comply with regulations, pass all scrutineering tests and demonstrate reliable on-track performance.
Pit Display	Must meet required dimensions, effectively engage viewers and incorporate relevant enterprise and engineering topics that represent our identity and journey.
Other Folios - Trade Brief, Compliance Booklet	Ensure all folios meet required formatting standards and comprehensively address all judging criteria.
Verbal Presentations	Verbal presentations must remain within time limits, address all key topics, engage judges through effective content and visual aids.
Renders	Renders must present realistic, visually appealing imagery while incorporating all major sponsor logos.

Overview

The Project Charter and Scope were instrumental in guiding execution, clearly defining objectives, boundaries and deliverables while strengthening team alignment, accountability, and structured project management.

Project Schedule & Time Management



Innovation. Precision. Success.



Figure 2 - Canva - Credit - Vivaan.R Velocity Vortex



Figure 3 - Canva - Credit - Vivaan.R Velocity Vortex



Figure 4 - Canva - Credit - Vivaan.R Velocity Vortex



Figure 5 - Canva - Credit - Vivaan.R Velocity Vortex



Figure 6 - Canva - Credit - Vivaan.R Velocity Vortex

Project Lifecycle

Our project timeline was established to define the major phases of development and provide the team with a clear understanding of project progression and overall direction.

Each phase was then deconstructed into a minimum of five key objectives (as seen in images on LHS) to ensure clarity, measurable progress, and structured execution. Completion of these objectives signified readiness to transition into the next phase, reinforcing controlled advancement.

Phase Objectives

The **Initiation Phase** focused on establishing formal foundations and defining project scope. The **Planning Phase** converted this foundation into structured strategies, resource allocation, and defined deliverables.

Although execution commenced early, a dedicated planning phase was intentionally implemented to strengthen preparation and reduce risk.

Monitoring and controlling measures were introduced as soon as major deliverables were underway, enabling early evaluation and minimising compromising on quality. This structured lifecycle approach ensured steady progress which led to a well prepared transition into the **Closure Phase** and competition readiness.

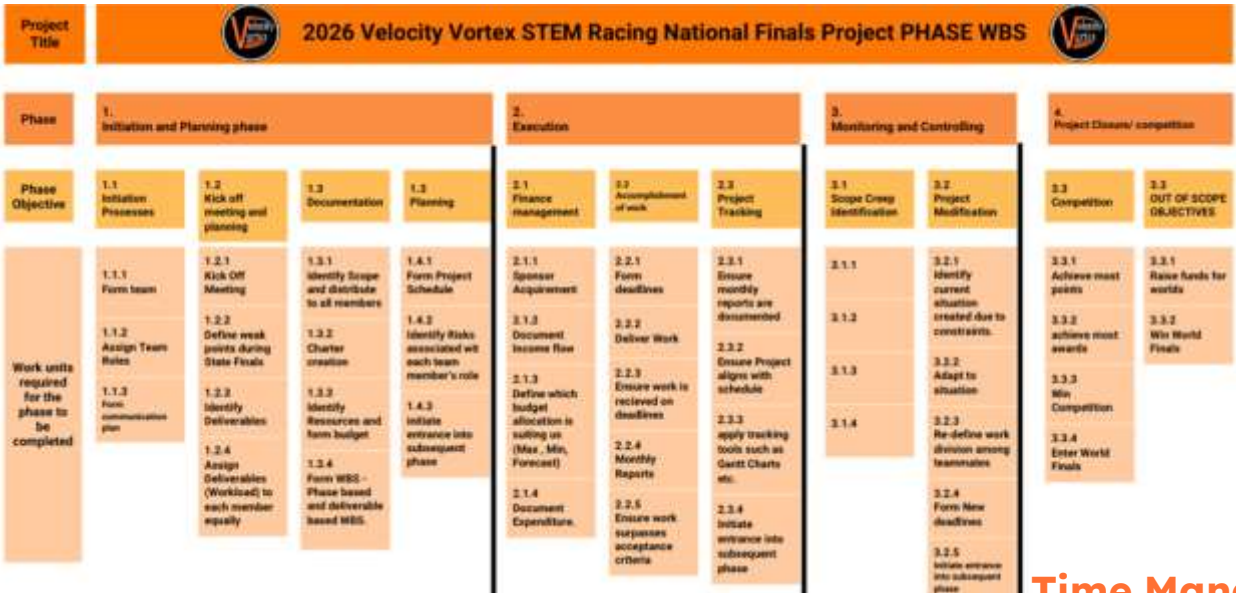


Figure 7 - Phase WBS - Canva - Credit, Vivaan.R Velocity Vortex



Figure 8 - Deliverable focused WBS - credit Vivaan. R, Velocity Vortex

Deliverable WBS

Each Major deliverable, i.e 'Trade Display' was broken down into the major steps which would be carried out to fully form the deliverable, i.e 'Design' & 'Building'. These steps were broken down into smaller work units which provided higher clarity to each team member regarding the estimated process which should be followed while undergoing the creation of their assigned major deliverable.

Gantt Chart - Time Management Techniques

The Gantt chart was an essential tool for planning and tracking project progress. By providing task timelines, dependencies, and estimations, it enabled precise deadline setting and efficient workload distribution. When the Gantt chart was integrated with our WBS and project lifecycle, it ensured coordinated execution of project elements, supported proactive monitoring, and strengthened overall team productivity, which directly contributed to the success of our project.

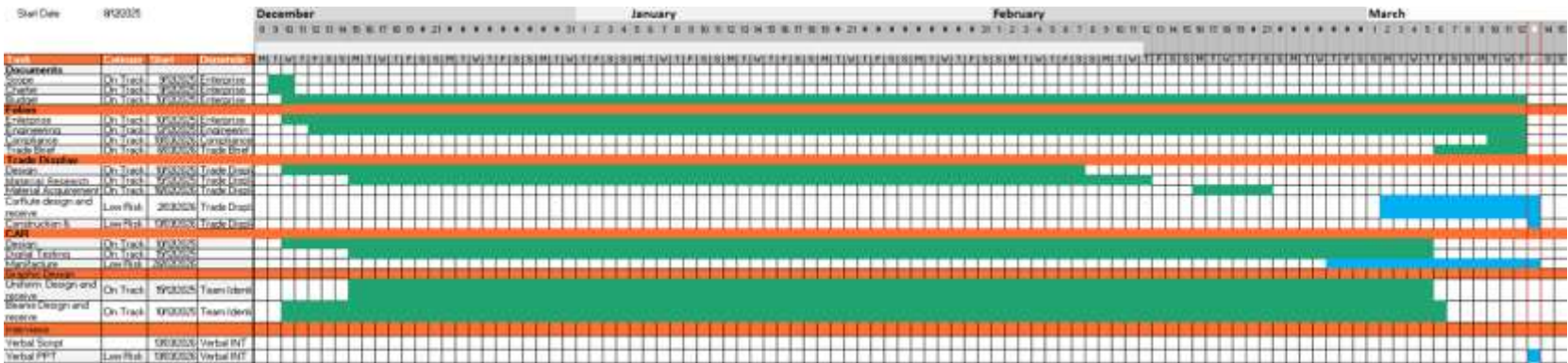


Figure 11 - Gantt Chart - Excel - Credit Vivaan.R, Velocity Vortex

Key - Gantt Chart
= Task Complete
= Task Ongoing

Work Breakdown Structures

To ensure structured planning and clear accountability, we development two Work Breakdown Structures (WBS) during the planning phase. A **Phase-Based WBS** outlined key project stages, objectives, and required processes (Figure 7), while a **Deliverable-Based WBS** defined each major output and its supporting components (Figure 8). Figure 9 presents the hierarchical structure of both models, with level titles adapted to suit their purpose. The implementation of dual WBS frameworks strengthened task clarity, coordination, and overall project control.

Figure 10, Canva - Credit, Vivaan. R

Time Management Techniques

Balancing competition preparation alongside our academic commitments required a structured and disciplined approach to time management. To ensure all deliverables were completed on schedule and to the highest standard, we implemented targeted planning strategies.

Deadline and Milestone Setting

Clear deadlines and monthly milestones were established to maintain momentum and accountability. These structured targets enabled incremental progress toward major deliverables and reduced last-minute pressure. Following the State Final, the school holiday period was strategically utilised to accelerate high priority tasks. Figure 11 demonstrates our December planning meeting, where monthly objectives were reviewed and allocated to ensure continued progress.

Structured Weekly Meetings

Weekly team meetings were conducted to maintain consistent progress throughout the project lifecycle. Each session followed a structured format: reviewing completed work, identifying outstanding tasks, and assigning objectives for the upcoming week. This continuous review cycle strengthened accountability, ensured equitable workload distribution, and supported steady advancement while balancing school work.

Overview

Structured deadlines, and weekly reviews improved productivity, reduced rescheduling conflicts, and enhanced overall deliverable quality. By managing time versus reacting to pressure, we maintained consistency, met benchmarks and strengthened our project management capability.

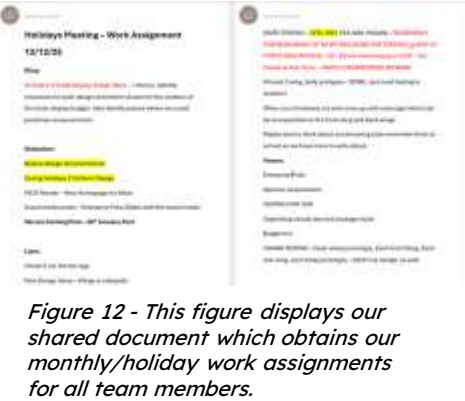


Figure 12 - This figure displays our shared document which obtains our monthly/holiday work assignments for all team members.

Budget & Resource Management



Innovation. Precision. Success.

Resource Management

Effective resource management was critical to project success. All resources were identified, evaluated, and acquired through a structured process aligned with our approved budget and performance objectives.

Before procurement, potential materials and services were researched, compared, and evaluated for cost, quality, and suitability. Where possible, products were tested or reviewed prior to final selection to ensure optimal performance while maintaining financial responsibility.

Resource Identification

A comprehensive resource identification process was undertaken during the budgeting phase. The budget and resources were categorised into key areas including manufacturing, trade display, portfolios, uniforms, and competition requirements. Each category was itemised and costed through supplier research and prior competition experience. This functioned as a Resource Breakdown Structure, clearly defining what resources were required, when they were needed, and their projected cost.

Figure 12, RBS in budget

Final Resource Cycle

A structured acquisition cycle was implemented:

1. Identify required resource
2. Research and compare suppliers
3. Evaluate cost, quality, and delivery timeframe
4. Confirm budget alignment
5. Approve and purchase

This ensured resources were acquired efficiently, within budget, and at the appropriate stage of the project cycle, preventing unforeseen expenditure.



Figure 13 - Canva - Visual representation of Resource Acquisition process - Credit Vivaan. R, Velocity Vortex

Accounting Method - Cash Based Accounting

Prior to preparing financial documentation, research was conducted in consultation with mentors to determine the most appropriate accounting method. We implemented a **cash-based accounting system**, where:

- Income was recorded when funds were received
- Expenses were recorded when payments were made.

Although straightforward, this method provided clear visibility of real-time cash flow and enabled accurate tracking of income and expenditure throughout the project. Regular updates to our financial records ensured that spending remained within approved limits and aligned with budget projections.

Budget

Using cash-based accounting, we developed a comprehensive budget to track expenditure and manage financial flow throughout the project. The budget was divided into four major categories: Car, Merchandise, Competition Requirements and Accommodation, and Trade Display. This structure allowed us to identify required resources, estimate costs, and monitor cash outflow, including the date each expense occurred, ensuring accurate and transparent tracking.

Velocity Vortex - Income Flow National Finals									
Income	Package	Tax (GST) optional		Gross Amount	Business Name	Date	Invoice/Rec'd	Status	
\$200.00	Supporter	\$0.00		\$200.00	Hon. John Gardner	28/01/2026	N/A	Income Received	
\$80.00	Supporter	\$0.00		\$80.00	Beanie Income From States	28-30/11/25	N/A	Income Received	
\$280.00	Visionary	\$0.00		\$280.00	Rajpal Family	6/02/2026	N/A	Income Received	
\$200.00	Supporter	\$0.00		\$200.00	Mrs. Costa (Modbury High School)		N/A	Income Received	
\$555.00	Innovator	\$50.45		\$504.55	North East Motor Repairs		SENT	Income Received	
\$550.00	Innovator	\$50.00		\$500.00	Lions Club		SENT	Income Received	
\$2,000.00	Elite	\$0.00		\$2,000.00	Modbury High School		N/A	Income Received	
\$672.00	Innovator	\$0.00		\$672.00	CASUAL DAY		N/A	Income Received	
\$1,600.00	Elite	\$145.45		\$1,454.55	Trojan	18/02/2026	SENT	Income Received	
Net Income		Total Tax (11%)		Gross Income					
\$6,137.00		\$246		\$5,891					

Figure 14A- Income Flow Document - Excel - Credit Vivaan.R Velocity Vortex

FINAL OVERALL BUDGET - Velocity Vortex 2026 National Finals									
Category and Item	Forecast/Budget	Allocated Maximum Budget	Allocated Minimum Budget	Actual Expenditure	Expected Date of Purchase	Date of Purchase	Variance	Outcome	Comments
Car -									
Bearings	\$ 150.00	\$ 200.00	\$ 100.00	\$ 120.38	20/02/2026	20/02/2026	\$ 29.62	Favourable	Due to shipping rates
WHEEL - PER 4	\$ 280.00	\$ 250.00	\$ 100.00	\$ 2.50			\$ 277.50	Favourable	Available at School - 3D PRINTED
Stickers	\$ 15.00	\$ 20.00	\$ 15.00	TBC	13/03/2026			TBC	
Paint	\$ 20.00	\$ 30.00	\$ 15.00	\$ 100.00	10/03/2026	10/03/2026	-\$ 80.00	Unfavourable	We needed more than we first thought
Primer	\$ 15.00	\$ 20.00	\$ 15.00	\$ 40.00	10/03/2026	10/03/2026	-\$ 25.00	Unfavourable	Our allocation at school went missing
Tether Guides	\$ 4.10	\$ 10.00	\$ 4.00	Available at School				Favourable	
Competition Requirements									
Entrance Fee	\$ 1,600.00	\$ 1,600.00	\$ 1,600.00	\$ 1,600.00	28/02/2026	28/02/2026	\$ -	Favourable	
Folios	\$ 100.00	\$ 150.00	\$ 100.00		13/03/2026	TBC	-\$ 100.00	TBC	
Merch and Graphics									
Uniforms	\$ 250.00	\$ 300.00	\$ 250.00	\$ 280.55	1/02/2026	6/02/2026	\$ 30.55	Unfavourable	Delayed due to design delays at Paladin
Logo Keychains									
Beanies	\$ -	\$ -	\$ -	\$ -			\$ -		SPONSORED BY RICHARDS FAMILY
Trade Display									
Metal for Backwall Frame	\$ 50.00	\$ 75.00	\$ 50.00	REPURPOSED FROM LAST YEAR				Favourable	Available at School
Metal for Shelves	\$ 80.00	\$ 100.00	\$ 70.00	\$ 75.00			-\$ 5.00	Favourable	
Corflute - Back & side, shelves	\$ 500.00	\$ 650.00	\$ 400.00	\$ 426.00	6/03/2026	10/03/2026	-\$ 74.00	Favourable	Delayed due to not having all members' images
Trestle Table			Available at School					Favourable	
Velcro - 3mm	\$ 15.00	\$ 15.00	\$ 10.00				-\$ 15.00	Favourable	
Packaging - Boxes	\$ 50.00	\$ 50.00	\$ 40.00					Favourable	
Accommodation + Food costs									
Food - Nearby Mentone Stores	\$ 250.00	\$ 200.00	\$ 300.00	TBC				TBC	
AIR BNB IN MENTONE	\$ 1,300.00	\$ 1,500.00	\$ 1,200.00	\$ 1,478.78	18/02/2026	18/02/2026		Unfavourable	
TOTAL	\$ 4,679.10	\$ 5,170.00	\$ 2,769.00	\$ 4,123.21				\$ 38.67	
Gross Income	\$ 5,891.00	Actual Expenditure	\$ 4,123.21	Difference	\$ 1,767.79				

Figure 14B - Budget - Excel - Credit Vivaan.R Velocity Vortex

Team Information & Roles

Velocity Vortex comprises four Year 11 students from Modbury High School, each assigned defined roles to ensure accountability, efficiency, and high performance.

Vivaan Rajpal - Team Manager

Job Function: Oversee team operations and dynamics, coordinate cross-functional collaboration, and manage the team’s financial strategy.
Responsibilities: Lead overall project coordination, monitor milestone progression, manage financial records, oversee sponsor relations, and develop the Enterprise and Compliance Folios.
Experience: Selected for his demonstrated leadership in sport and entrepreneurship programs, showcasing strong communication, organisation, and strategic oversight capabilities.



Photo Credit - Velocity Vortex

Liam Bairstow - Lead Engineer

Job Function: Design, manufacture, refine and test a fully compliant, high-performance car.
Responsibilities: Develop and manage the Engineering folio, lead CAD and CAM modelling to design and manufacture the car, conduct CFD and track testing, and design performance-enhancing components and accessories.
Experience: Chosen for his proficiency in Fusion 360 and CFD analysis, recognised through receiving the “Best Designed Car” award.



Photo Credit - Velocity Vortex

Sebastian Richards - Graphic Designer & Social Media Manager

Job Function: Maintain a cohesive and professional brand identity across platforms.
Responsibilities: Deliver all visual WBS outputs including logo design, uniforms, portfolio graphics, and promotional materials, while managing social media strategy and engagement.
Experience: Brings experience in Photoshop, digital and aviation photography, and brand development, strengthening the team’s visual identity and online presence.



Photo Credit - Velocity Vortex

Riley Hutchinson - Website & Trade Display Manager

Job Function: Develop and manage the Trade Display and digital presence to enhance sponsor visibility and engagement.
Responsibilities: Oversee Trade Display design and electronic systems, maintain the team website, and support sponsor management and communication, ensuring physical and digital deliverables align with project milestones.
Experience: Selected for technical proficiency in CAD software and programming with HTML, CSS, and JavaScript, ensuring effective representation of the team through both the Trade Display and our website.



Photo Credit - Velocity Vortex

Team Interactions

Each team member was assigned a specific role to ensure clear communication channels and promote effective collaboration across all tasks. This structure ensures that every team member understands their responsibilities and how their contributions support the overall project goals. A communication diagram was developed to illustrate how seamless interaction supports efficient problem-solving. Additionally, a responsibilities table was created to maintain accountability, clarify roles, and strengthen teamwork, and internal communication.

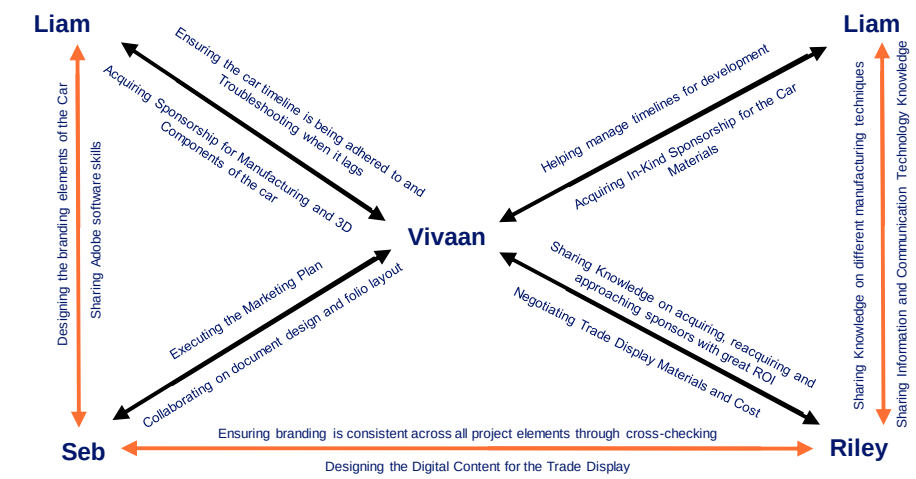


Figure 15 - Team Interactions - Google Slides - Credit Vivaan.R, Velocity Vortex

Team Tasks & RACI Matrix

As shown in Figure 15, team roles and responsibilities were clearly structured through a defined communication flow, enabling each member to identify the appropriate person to consult on specific tasks. The RACI Matrix illustrates how responsibility, accountability and communication were managed throughout the project, ensuring efficient collaboration and informed decision making.

2026 VELOCITY VORTEX NATIONAL FINALS RACI MATRIX				
	Team Leader Vivaan Rajpal	Lead Engineer Liam Bairstow	Graphics Designer Sebastian Richards	Trade Display Manager Riley Hutchinson
Car: Physical appearance	Consulted	Responsible	Responsible	Consulted
Car: Engineering	Assist	Responsible	Informed about	Informed about
Car: Manufacturing	Assist	Responsible	Informed about	Informed about
Car: Testing (on track)	Responsible	Responsible	Assist	Assist
Car: Testing (CFD/VWT)	Responsible	Assist	Informed about	Informed about
Sponsor Acquisition + Sponsor Management	Responsible	Consulted	Consulted	Assist
Engineering Folio	Responsible	Responsible	Informed about	Informed about
Enterprise Folio	Responsible	Consulted	Responsible	Responsible
Compliance Folio	Responsible	Assist	Informed about	Informed about
Uniform Design	Assist	Assist	Responsible	Assist
Beanie Design	Assist	Assist	Responsible	Assist
Trade display/Trade Display Content	Assist	Assist	Responsible	Responsible
Trade Display Frame	Assist	Consulted	Consulted	Responsible
Trade Display Brief	Consulted	Informed about	Consulted	Responsible
Trade Display Evolution	Assist	Assist	Informed about	Responsible
Verbal Script	Responsible	Responsible	Responsible	Responsible
Verbal PPT	Responsible	Responsible	Responsible	Responsible
Social Media	Assist	Consulted	Responsible	Consulted
Team Branding	Responsible	Informed about	Responsible	Informed about
Upcoming Deadlines/milestones	Responsible	Responsible	Responsible	Responsible

Figure 16 RACI Matrix - Excel - Credit - Vivaan.R, Velocity Vortex

Communication Challenges and Solutions

Balancing academic commitments alongside the demands of the STEM Racing competition presented challenges in maintaining consistent communication and meeting deadlines. This became more significant following the SA State Final when school holidays began, as the team no longer had regular access to meet within the school environment. To overcome this, we organised regular holiday meetings at each other’s homes to ensure continued progress, as pictured in Figure 17.



Figure 17 - Holiday meetings at each other's houses - credit - Velocity Vortex

Communication Methods Between the Team

Effective communication was essential to maintaining a collaborative and productive team environment. It allowed members to share ideas, coordinate tasks and ensure the project progressed efficiently. To support this, we implemented several communication methods, outlined below.

Group Chat - Whatsapp

Whatsapp proved to be an essential communication platform as it allowed team members to stay connected during holidays and weekends, regardless of their location, while also providing a platform to share updates, set deadlines and discuss upcoming milestones. Alongside these benefits, it also provided a channel for our lead teacher to offer support and guidance while overseeing our progress. Additionally, the chat was used for practical coordination, such as confirming uniform sizes and logo preferences.



Figure 18a - Usage of Whatsapp, Credit - Velocity Vortex

Phone Calls

Phone calls were used to communicate within the team and with sponsors, providing real-time discussions and updates similar to face-to-face meetings.

Google Meet

Google Meet was a key communication tool, particularly during school holidays when the team could not meet in person. While WhatsApp was primarily used for quick updates and informal communication, Google Meet enabled structured discussions that closely replicated our face-to-face meetings. This allowed the team to collaboratively review deadlines, allocate tasks and make informed decisions regarding upcoming milestones, ensuring the project remained organised and on schedule.

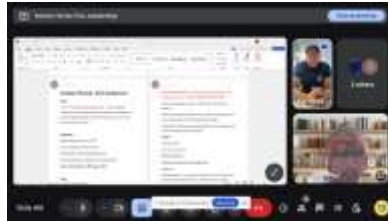


Figure 18b - Usage of Google Meet, Credit - Velocity Vortex

Face-to-face communication

Face-to-face, daily communication at school provided the most direct and effective way for our team to communicate. This allowed us to discuss current issues and collaborate on solutions in real-time.

Overview

WhatsApp enabled quick updates and informal chats, face-to-face meetings allowed real-time problem solving, Google Meet enabled structured discussions, while phone calls provided immediate discussions when needed.

Sponsorship Strategy

Sponsorship acquisition was approached as a structured, conversion focussed process instead of informal outreach. Building on experience from the State Final, we refined our engagement model to increase our professionalism, clarity and sponsor confidence. Target businesses were identified based on local engagement, industry alignment, and possible mutual benefit. Each sponsor was presented with our prospectus outlining sponsorship packages, defined ROI, and brand exposure opportunities.

Four-Step Sponsorship Acquisition Plan

To ensure consistency and accountability, a four-step acquisition process was implemented:

- 1. Identification & Initial Outreach: Targeted businesses were researched and contacted via an email, accompanied by a detailed sponsorship prospectus.
- 1. Formal Meeting/Discussion (In-person, Phone Call, or Online) A formal discussion was conducted to present our value proposition, outline sponsorship packages, and address questions directly, which strengthened our credibility and trust.
- 1. Follow-Up & Confirmation: Structured follow-up phone calls ensured clarity, reinforced interest, and moved sponsors toward commitment.
- 1. Invoice & Transaction Management: Once confirmed, formal invoices were created and transactions recorded using cash-based accounting, ensuring financial accuracy.



Figure-23 - Sponsorship Plan - Canva Credit, Vivaan. R, Velocity Vortex

ROI Suite

Our ROI suite ensured sponsors and stakeholders received measurable value while allowing us to manage the project in a professional, real-world business environment, strengthening our future-ready skills. Packages are listed below.

Velocity Vortex Sponsorship Packages		
Package name	Donation Range	Package ROI List
Elite	\$1001+	- Large logo featured on team uniform - Dedicated social media recognition (business feature or team visit) - Large logo on the race car (or two standard logos depending on space availability) - Innovator, Visionary and Supporter benefits
Innovator	\$501 - \$1000	- Acknowledgement in interview and speeches - All Visionary and Supporter benefits
Visionary	\$251 - \$500	- Logo displayed on the back of the team uniform (larger than Supporter tier) - Larger logo featured on the Trade Display - All Supporter benefits
Supporter	\$50 - \$250	- Social media announcement featuring the business - Logo on the front cover of team portfolios - Small logo on the race car - Trade Display and team uniform

Figure 24, Sponsorship Packages, Canva - Credit Vivaan.R, Velocity Vortex

Uniforms

Team uniforms served as high-exposure sponsorship assets for our registered internal and external stakeholders due to their visibility during judging, presentations, and networking sessions. We developed a unique and innovative ideology for logo placement for the National final shirts which followed a proportional hierarchy, with size and positioning aligned directly to sponsorship tier. This is demonstrated below:

School Logo Placement - Sponsor

The school logo was placed prominently on the shoulders of our polo shirts, recognising both significant financial support (\$2200) and entrepreneurial opportunities. This placement ensured strong visibility while reinforcing partnership and ongoing support of staff, teachers, and previous competing STEM Racing students.

Major Sponsor Logo Placement

Primary sponsor logos were positioned higher on the shirt and scaled in proportion to their contribution level. For example, Trojan Aluminium's \$1600 sponsorship was represented through increased logo size, ensuring a return on their investment which displayed our gratitude.



Company and Supporter logos

Logos of supporting companies who contributed materials or services (e.g., Paladin Sports and Boca Bearings) were incorporated in designated areas, recognising their contributions while again maintaining visual hierarchy and brand consistency.

Car - Logo Placement

The competition car was prioritised as a primary sponsorship asset due to its central role and importance within the competition. Sponsor logos were strategically positioned on the car to maximise visibility during scrutineering, racing, and when on display. Logo placement was structured according to sponsorship tier, forming a clear return-on-investment (ROI) system. This tiered approach ensured transparency, fairness, and sponsor recognition while reinforcing the direct link between financial support and brand visibility.

Canister Logos

Logos placed on the canister housing occupy the largest surface area on the car. This premium placement was for major sponsors and reflected their level of contribution and support.

Side Pod Logos

The side pods feature the STEM Racing governing body and our school logo. This prominent position on each side of the car ensures high visibility, compliance and partnership representation.

Front Wing Logos

Logo placement followed a structural hierarchy, with size and positioning proportionate to sponsorship contribution. The front wing provided a secondary branding location for supporting sponsors while keeping visual balance. This tiered approach ensured fair ROI across all sponsorship levels.



Stakeholder Register

The stakeholder register was an essential part project management tool that identified all individuals and organisations with an interest in our project. It documented stakeholder roles, contact details, level of influence, and communication methods & frequency, to ensure no key relationship is overlooked. Maintaining the register strengthened accountability, supported targeted communication and enabled us to prioritise high-impact stakeholders such as sponsors and mentors. The register also outlines the type and frequency of communication used with each stakeholder.

Name	Contact/Organisation	Role	Category	Contact	Form of Communication	Communication Frequency
John Gardner	SA Liberal Party	Sponsor	External	Phone: (08) 8380 4234 Email: John.Gardner@parliament.sa.gov.au	Phone Calls, Newsletters, & Email	Newsletters sent monthly, Emails sent on a need basis
Joanne Costa	Modbury High School	Sponsor	Internal	Email: joanne.costa123@schools.sa.edu.au Phone: withheld for privacy	Personal Meetings, Emails, Daymap	Communication remained frequent due to meetings at school - Weekly or Fortnightly
Scott Posingham	North East Motor Repairs	Sponsor	Internal	Phone: 8402 434 853 Email: Scott.posingham@gmail.com	Phone Calls, Video Conferences, Newsletters, & Email	Newsletters sent Monthly, Messages sent on a need basis
John Chellachis	U/ONS Club of Adelaide Metrists	Sponsor	Internal	Phone: N/A Email: N/A	Newsletters and Email	Monthly (Newsletters)
Michelle Aravaz	Trojan Aluminium	Sponsor	Internal	Phone: N/A Email: mjan@trojan.com.au	Newsletters and Email	Monthly (Newsletters)
Karl Read	Modbury High School	Mentor - Supervising teacher	Internal	Phone: 8484 337 008 Email: karl.read297@schools.sa.edu.au	Personal Meetings, Emails, Daymap	Daily communication through school and technology
Mary Anne Richards	Mary Anne's Hair Salon (Fundraiser's Co-Ordinator)	Mentor - Sponsor Acquisition	Internal	Phone: withheld for privacy Email: mrichards1975@gmail.com	Personal Meetings, Emails, Messages	Regularly
Deepak Rajpal	Withheld for Privacy	Mentor - Financial Advice and Accountability	Internal	Phone: withheld for privacy Email: de.rajpal@gmail.com	Personal Meetings	Communication remained frequent on a need basis
Stephen Read	STEM Racing Australia	Mentor - Engineering	Internal	Phone: withheld for privacy Email: stephenread93@gmail.com	Phone calls & Messages	Calls were made weekly or fortnightly to receive feedback
James Coulter	JE Designs	Mentor - Graphic Design	Internal	Phone: Withheld for Privacy Email: helio@couturedesigns.com.au	Email	Fortnightly - Check-in & Feedback
Mason Green	AUS RAMD - Fitter and Turner	Mentor - Engineering + Trade Display	Internal	Phone: withheld for privacy Email: Maseo.green04@gmail.com	Messages and phone	As needs basis - conversation and feedback

Figure 25 - Excel - Stakeholder Register - Credit Vivaan. R, Velocity Vortex

Stakeholder Communication and Engagement

Stakeholder communication remained a strategic priority to ensure transparency, accountability, and continued engagement. Sponsors received monthly newsletters providing progress updates, key milestones, and recognition of their contribution. These updates highlighted major developments, including engineering advancements and merchandise releases such as our beanies, reinforcing the impact and visibility of their support.

To strengthen our sponsor engagement beyond reporting, our newsletters incorporated interactive elements such as a Formula-1 Racing or Velocity Vortex themed Wordle, fostering a sense of fun, involvement and partnership rather than just sponsorship. Similarly, sponsors were presented with a team beanie and Certificate of Sponsorship as a tangible symbol of appreciation and partnership. The beanie was a core element of our team identity, reinforced inclusion within our culture, while the certificate provided professional recognition of their investment.

Overview

This consistent and value driven communication approach strengthened sponsor relationships, reinforced return on investment, and contributed to long-term stakeholder trust.



Figure 26, 1st Velocity Vortex Newsletter - Canva, credit - Vivaan. R, Velocity Vortex



Figure 27, Velocity Vortex custom Wordle - Credit - Velocity Vortex



Figure 28, Certificate of Sponsorship



Team Branding

Team Name and Slogan

The name Velocity Vortex was strategically selected to embody the team’s commitment to speed, intensity, and controlled performance. It reflects our focus on delivering high-speed outcomes with precision and purpose. To reinforce this identity, a clear and purposeful slogan was developed around three key values. **Innovation** represents the team’s creative problem-solving and ability to differentiate itself. **Precision** reflects attention to detail across engineering, enterprise, and presentation elements. **Success** represents our results-driven mindset, transforming innovation and precision into measurable performance outcomes. Ultimately, our team name and logo go beyond a label, embodying the bold and driven identity of Velocity Vortex.

Team Logo Development

During our kick-off meeting, we determined that a full rebrand of our existing logo was unnecessary. The logo was already clear, recognisable and aligned with our team identity. Instead, we strategically enhanced it through the integration of our team slogan, strengthening its communicative power while maintaining strong brand consistency. Pictured below is the evolution of our logo:

State Final Logo
This was the logo which was used during the South Australian State Final.

Prot. 1
Eventually, we chose to apply our slogan through a simple and attractive approach

Prot. 2
This experimental approach explored innovation yet remained too complicated

Prot. 3
This approach allowed us to combine simplicity and innovation.

Prot. 4
Returning to simplicity we applied our slogan above and below the team name.

Final Design
Our final design presents a bold logo which incorporated our simple yet perfect logo along with our slogan!

Final Logo Design and Rationale

Logo Circular Shape:

The circular shape was intentionally selected to maximise versatility and scalability. Its balanced geometry allows seamless application across multiple platforms and formats - including uniforms, merchandise, digital media, documentation and car livery - without compromising clarity or visual impact. The contained shape also reinforces unity and cohesion within the team.

Sharp Font:

A sharp, clean font was applied to reflect precision, speed and professionalism. The angular characteristics of the font convey forward momentum, aligning with the high-performance nature of STEM Racing while ensuring strong legibility across both print and digital applications.



Large Slanted V:

The prominent slanted “V” forms the visual focal point of the logo and directly represents Velocity Vortex. Its straight edges symbolise strength, stability and technical precision, while the angled orientation communicates motion and acceleration. The subtle curvature at the top conveys the aerodynamic flow and movement of an F1 car at speed.

Curved Track Lines:

The three curved lines positioned below our team name represent a 2D racing track. Their flowing form reinforces the concept of speed, trajectory and continuous progression, visually representing the meaning of “Velocity”. These lines add movement to the composition, preventing the logo from appearing static.

Beanie Design & Merchandise Strategy

Our beanie was developed as a bold and highly visible extension of our team identity. As an original merchandise piece within the competition environment, it allowed us to differentiate ourselves while reinforcing brand recognition. The design incorporates our core colour palette - Bright Orange, Eerie Black and White - ensuring immediate visual alignment with our established branding. The white team name across the crown enhances visibility, while the orange cuff positions our logo prominently at eye level, maximising exposure in both formal and informal settings.

The initial prototype, designed by Sebastian (Figure 19) in collaboration with the team, explored a more complex graphic integration. However, due to production constraints, we strategically returned to our original concept (Figure 20). This decision allowed us to deliver, simplicity and strong visual impact - ensuring delivery before the competition without compromising brand quality. With winter approaching and strong football culture across most States in Australia, beanies are both practical and highly visible. Their frequent use in outdoor environments increases brand exposure, strengthening recognition of the Velocity Vortex identity beyond the competition space. By designing our beanie with purpose and practicality in mind, we transformed it to function not only as merchandise, but as a strategic branding asset.

Font Choices

To maintain consistency across our team branding we established font choices and sizes for specific locations across the folios as listed below:
Front cover of folios and Banner - **Lexend Normal 28 Bold**
Page Topic Headings - **Lexend Normal 15 Bold**
Subheadings - **Lexend Normal 12 Bold**
Basic Text - Lexend Normal 10
Text within Boxes - Lexend Normal 8.5/6.5
Image References - *Lexend Normal Italics - 8*

Team Colours

Our team colour palette was strategically selected to create a bold, recognisable and cohesive visual identity across all brand assets. **Bright Orange** represents our energy, innovation and competitive drive. As a high visibility colour, it ensures strong brand recognition in competition environments and digital platforms, allowing our team to stand out instantly. **Eerie Black** symbolises precision, performance and our ambition to push boundaries. Inspired by the night sky, it reflects our aspiration to reach for the stars while adding depth and contrast to our visual identity. **White** reinforces clarity, professionalism and focus. It enhances readability across portfolios, uniforms and trade display materials, ensuring our branding remains clean, sharp and accessible.

Primary Colours

Bright Orange
#ff7003ff

Eerie Black
#1f1f1fff

White
#fffffffff

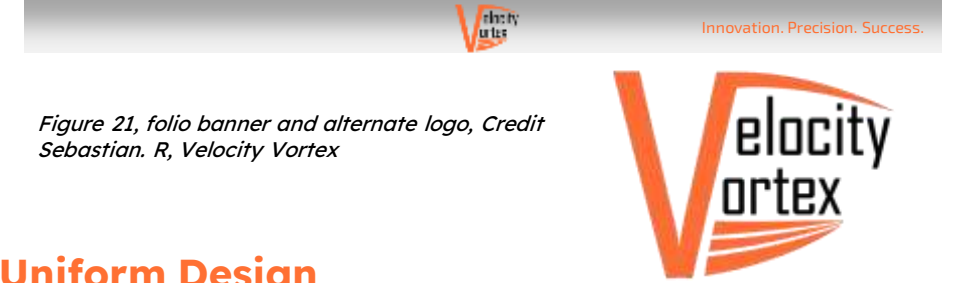
Amber
#ffab40ff

Medium Grey
#666666ff

Silver Grey
#d9d9d9ff

Banner

Our banner was strategically developed as a unifying visual device across all team documentation and communications. The fade between grey and dark grey symbolises simplicity and complexity - reflecting our journey from concept to competition and our continuous pursuit of excellence. The gradient effect introduces depth and modernity, reinforcing our innovative and forward-thinking identity. It was used across our portfolios, online posts, and on invitations and correspondence for stakeholder meetings, ensuring consistent and professional branding. As can be seen in Figure 21, a refined logo variation was applied to our banner to ensure strong contrast across all backgrounds, with the logo and slogan **intentionally** separated to enhance layout flexibility, demonstrating its versatility across multiple branding applications..



Uniform Design

Our uniforms were strategically designed to reinforce our professional identity while ensuring strong visual presence within the competition environment. A clean white polo was selected as the foundation to reflect clarity, professionalism and confidence. The neutral base enhances contrast, allowing our brand colours to remain bold and highly visible. Two Bright Orange V-shaped arrows were incorporated on the front and back of the shirt, directly referencing the central “V” in our logo. This creates immediate visual alignment between our uniform and overall brand identity. The angular design represents forward momentum, ambition and performance - reinforcing the direction and drive that defines Velocity Vortex.

Strategic Impact

By integrating consistent colour placement and logo alignment across our uniforms, car livery, trade display and merchandise, we strengthened brand consistency and ensured our team presents a unified and recognisable presence in both formal judging sessions and the less formal competition environments.



STEM Racing Logo

The STEM Racing logo was positioned on the upper right chest to reflect official competition alignment and ensure clear visibility in presentations.

Sponsor Logo Placement

Sponsor logo placement was strategically integrated to deliver clear ROI while enhancing the professionalism of our uniform design. The structured positioning strengthened visual balance and utilised negative space effectively, particularly across the back of the shirts.

Social Media Strategy (Planned Activation)

Due to government regulations restricting active social media use for individuals under the age of 16, our team was initially unable to maintain an active online presence. Developing a structured social media strategy remained an important component of our marketing and sponsor ROI. To ensure compliance, we planned our social media strategy in advance and scheduled content to be published once all team members reached the required age, with Sebastian turning 16 on 11th March. Our strategy focused on consistent and engaging content to highlight our team's journey. Throughout progress updates, milestone achievements and promotional posts, we aim to build a stronger community presence. Increase awareness of the competition and our team while sharing our journey with supporters. This ensured compliance while enabling a structured and effective social media presence.

Promotional Posts

Promotional posts were strategically planned to raise awareness of both the competition and our team while maintaining consistent communication across our social media platforms. These posts recognise our supporters while increasing their visibility within the wider community. Our first post announces our qualification for the National Final, followed by dedicated sponsor recognition posts highlighting their support. Once published, sponsors and key stakeholders are notified via email to ensure they are aware of the exposure and acknowledgement.

Progress Updates

From 13th March, when our social media presence becomes active, progress updates will provide followers with insight into our journey, while also serving as an alternative to our newsletter. These posts will showcase key milestones from both the competition and earlier project stages, including car development, uniform design, Trade Display construction, and portfolio production, while also documenting our travel to Melbourne, our experience at an AFL game in Melbourne, Trade Display setup, competition registration and progress throughout the National Final, allowing supporters to also follow our experience in real-time.

Together, these posts strengthen brand awareness while keeping sponsors and supporters engaged with our competition journey.



Figure 29 - Reference - Instagram - Team Promotion Post - Credit - Sebastian Richards

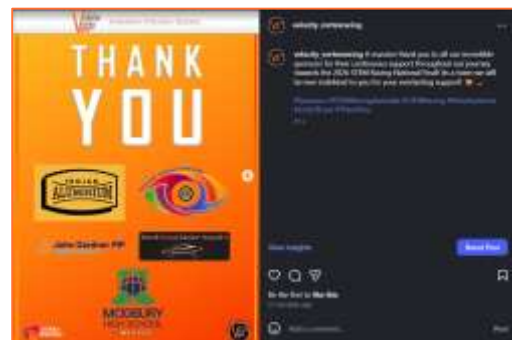


Figure 30 - Reference - Instagram - Sponsor Acknowledgement Post - Credit - Sebastian. Richards



Figure 31 - Reference - Instagram - Progress Update Post - Credit - Sebastian. Richards

School Community Events

School events, such as Modbury High School's Open Night, held in February, provided valuable opportunities to promote our team and increase awareness of the STEM Racing competition. Throughout the evening, we engaged with numerous attendees who expressed strong interest in both the competition and the STEM Racing program offered at the school. Our Team Leader, delivered detailed explanations of each component of the competition, helping visitors understand the scope of the program. The race track display was a key attraction, drawing parents and prospective students who were eager to experience the thrill of racing.

Promotion at Open Night

During Open Night, we actively promoted our team by directing attendees to our website. The site included a dedicated contact page, enabling visitors to submit enquiries or send emails regarding the competition or the STEM Racing subject. This facilitated effective communication and strengthened engagement with our school community.

Target Audience

Our social media strategy was developed to engage a broad yet relevant audience who support our team's visibility and growth. Key target groups include:

- Sponsors and Potential Sponsors:**

Industry partners who follow our updates to assess our progress, professionalism, and value as a team they may support.

- School Community:**

Current students, parents, and staff who engage with our posts to stay updated on our achievements and activities in STEM Racing.

- Prospective Students and Families:**

Our content aims to inspire future interest in the subject and competition.

- Competition Network:**

Judges, mentors, and other STEM Racing teams fostering collaboration,



Figure 32 - Liam and Jack May (STEM Racing Alumni) showcasing main features of the car and track. - Credit Sebastian. R



Figure 33 - Liam and Jack May (STEM Racing Alumni) handling the track for spectators allowing for a interactive experience which provides higher exposure to the team and the competition.

Open Night Exposure

Our engineer, Liam, captured audience attention as he demonstrated how the car launched down the track, showcasing the precision, speed, and engineering behind our car design. With the race track ready and a clear explanation of the competition, he guided the audience through the launch process, transforming a simple demonstration into an exciting insight into the world of STEM racing. His energy and expertise drew crowds, highlighting both our team's innovation and the thrill of the competition.

Suitable Platforms

In selecting our social platforms, we evaluated options that would maximise visibility and effectively reach our target audience. Instagram was identified as the most suitable due to its strong presence of F1 related content, including official teams, engineering communities, and student competitors. We employed targeted strategies such as industry-related hashtags to enhance content discoverability and ensure our posts appeared to audiences with aligned interests. This enabled us to engage with F1 enthusiasts, aspiring engineers, younger audiences, and key stakeholders. Instagram also serves as a central hub for many STEM Racing teams and sponsors, allowing us to build connections, strengthen professional relationships, and maintain consistent communication across the competition network. As a result, Instagram proved to be an ideal platform for promoting our team, increasing exposure, and expanding our digital engagement.

Media Exposure Alternative Techniques

Trade Display

Our Trade Display includes dedicated sections highlighting our digital presence, featuring a Linktree QR code. This allows visitors, including judges, other teams, and general audience, to instantly access our website and social media platforms. By integrating this interactive element, we increase engagement by allowing visitors to our display to explore additional content beyond the physical display, and provide greater exposure for our team across multiple digital platforms. Pictured opposite is our Linktree QR code.



Figure 34 - Linktree QR Code - Credit - Riley. Hutchinson, Velocity Vortex

Logo Application - Brand Exposure

Effective promotion of any brand relies on consistent use of its logo across all communications and deliverables. Following this principle, our logo was prominently applied across major posts, newsletters, sponsorship prospectuses, portfolios and key team deliverables. This consistent application enhanced brand visibility and recognition among our audience. Examples of our logo usage are shown below and to the right.

Figure 35- Logo exposure



Display Design, Development & Integration

The Velocity Vortex Trade Display was developed through a collaborative, iterative, and sustainability-focused process, driven by our commitment to efficiency, cost-effectiveness, compliance, and visual appeal. Our goal was to create a display that not only met the competition requirements but also surpassed our previous design in functionality, professionalism, and aesthetics. This project was an opportunity to apply the lessons learned from our State Final experience, ensuring that every aspect of the new design addressed past flaws while introducing innovative improvements.

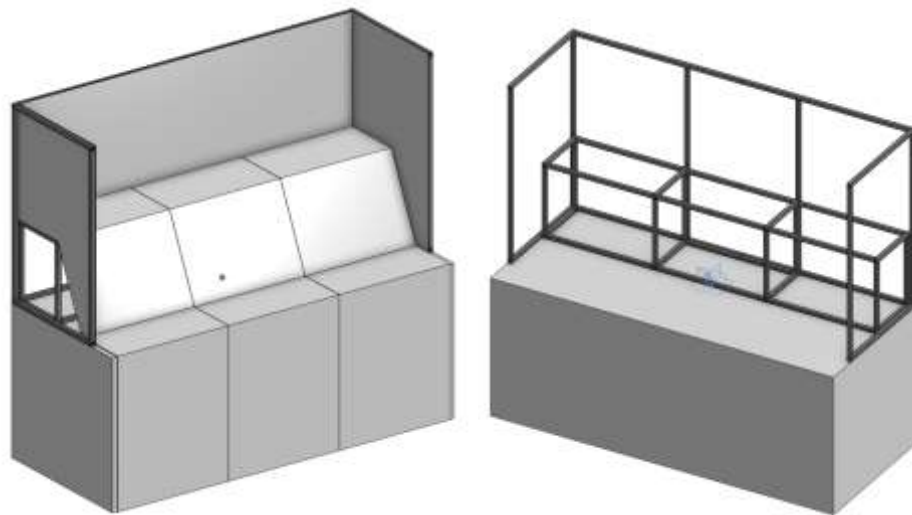


Figure 36 - CAD model with and without Corflute. Credit - Riley.H

Design Inspiration and Iterative Development

The design process began with a collective brainstorming session, where we critically analysed the weaknesses of our previous trade display. By identifying key areas for improvement, we aimed to create a design that was as close to perfect as practically possible. Riley took the lead in modelling the display using Onshape, a browser-based CAD tool that allowed the entire team to easily access and review iterations. Onshape was chosen not only for its collaborative features but also due to technical difficulties encountered with other software. Over the course of twenty iterations—three of which involved major redesigns—we refined the concept to meet our criteria for functionality, aesthetics, and practicality. The final design emerged after extensive team discussions, ensuring that every detail was carefully considered and optimised.

Display Layout: Showcasing Innovation

Initially, we planned to feature the car as the centrepiece of the display. However, as the design evolved, we realised that the shelf frame structure made this impractical. Instead, we focused on creating a clean and professional layout that effectively showcases the car while highlighting our team's achievements. The car is strategically positioned to draw attention, surrounded by information panels that detail our journey, team roles, and the development of the car itself. Our logo, sponsors, and a QR code linking to our social media and website are prominently displayed, adding a professional touch and encouraging engagement. The use of Corflute strips provides both structure and flexibility, allowing us to adjust the angle and layout of the shelves to best present our content in an engaging and easy to follow way.

Modularity and Adaptability

Modularity was a key consideration in the design, ensuring that the display could be easily assembled within the competition's strict 20-minute setup time limit. We conducted multiple setup tests to refine the process, guaranteeing that everything fits together seamlessly and efficiently. The display's adaptability is further enhanced by the Corflute material, which allows for adjustments in the layout while maintaining a cohesive appearance. By reusing the steel frame from our previous display, we not only promoted sustainability but also ensured the structure's durability and ease of assembly.

Compliance and Sustainability

Sustainability and compliance were at the heart of every design decision. We maximised resource efficiency by repurposing materials from our previous display, including the steel frame and a trestle table, significantly reducing both costs and environmental impact. While we considered more sustainable materials like Ecoboard, Corflute was ultimately chosen for its cost-effectiveness and familiarity, given the constraints of our budget and timeline. The steel frame, though not as lightweight as aluminium, was selected due to its availability in our school's Tech Studies facility. Additionally, we repurposed tablets from a previous team, further reducing e-waste and promoting resourcefulness. Compliance with competition regulations was also a priority, particularly in ensuring that the display could be packed and transported in standardised boxes without the need for complex packaging solutions. This material combination balanced functionality, durability, and visual presentation while supporting efficient assembly and transport.



Figure 37 - Renders of our State finals trade display design, which shows their design differences. Credit - Riley.H Sebastian.R

Aesthetic and Branding Choices

The display's colour scheme of black, white, and orange was carefully chosen for its striking visual impact and versatility. Black and white provide a clean, neutral backdrop that allows the vibrant orange accents to stand out, drawing attention to key elements such as the car, logo, and informational panels. The Velocity Vortex name, logo, and slogan are integrated throughout the display, reinforcing our team's identity—speed, precision, innovation, and success. This cohesive branding creates a unified and professional appearance that aligns with our team's values and enhances the overall visual appeal of the display.

Team Collaboration, Budget Management, and Lessons Learned

The development of the Trade Display was a truly collaborative effort. Riley led the CAD modelling process, ensuring that the design was both functional and visually engaging, while also assisting in sourcing materials like Corflute. Vivaan oversaw material procurement and budgeting, coordinating with other teams to repurpose old components and working with the Tech Studies division to secure necessary equipment and materials. Assembly was a team effort, providing all members with valuable experience in time management, adaptability, and quick decision-making when challenges arose.

Budget

Initially, we anticipated tight budget constraints for the Trade Display. However, through careful material selection and repurposing components from previous teams and our State Final display, we delivered the display well under-budget, demonstrating strong financial planning and resource efficiency. For an in-depth budget breakdown refer to our Budget and Resource Management slide on page 4.

Final Outcome: A Display That Represents Our Journey

Our Trade Display is more than just a competition requirement—it is a reflection of the creativity, dedication, and hard work of every team member. It tells the story of our journey, from the initial car prototypes to our experiences at the State Final and beyond. Through thoughtful design choices, cost-effective materials, and a strong emphasis on sustainability, we have created a display that is not only compliant and functional but also visually stunning. We are incredibly proud of the final result, which stands as a testament to our team's innovation, spirit, and commitment to excellence. This display represents who we are as a team, and we are excited to present it as a showcase of our achievements.



Figure 38- Panel Designs for the trade display. Credit - Riley.H, Sebastian.R, Vivaan.R, Liam.B

Risk Management

Risk identification was a key component of the project, enabling us to prepare for a range of scenarios and address potential obstacles proactively. During the project initiation, each team member identified risks relevant to their role, which formed the development of our risk table and associated mitigation actions. This documentation provided a structured analysis of the possible risks, and supported the creation of effective solutions and preventative measures to minimise their impact throughout the project.

Risk Table

The Risk table allowed us to address three key areas of risk management: preventative controls, mitigation strategies, and the potential impact on project resources, timing, scope and quality. Preventative controls outlined the actions required to minimise or eliminate each risk before it occurred, while mitigation strategies detailed the steps to be taken should the risk eventuate. Additionally, the table provided a clear assessment of how each risk could affect project constraints and deliverable quality, ensuring the team was well prepared to respond effectively.

Risk Matrix

The Responsibility Assignment (Risk Matrix) table on the RHS outlines the actions to be taken if a risk materialises, categorised into three levels: Extreme, Moderate and Low. This structure ensured all team members understood the required response for each risk level, providing clear guidance on how to proceed in the event of an issue.

Monthly Status Reports - 4/02/26 -> 28/02/26 - Velocity Vortex; National Finals		
Team Member	Completed Work	Incomplete Work/WORK IN PROGRESS
Overall Team Combined Complete and incomplete work	• Car Name Decided • Accomodation Finalised • Flight Plan and travel Finalised - Who's doing what • Casual Day Done - 672	Verbal Scripts & Presentations
Vivaan	• Uniforms Paid For • Budget Done • Newsletters	• Compliance Folio • Testing Final car, wheels • Enterprise Folio
Sebastian	• Uniforms Ordered and Paid • Beanies Design Confirmed and Sent • Final Logo + Slogan	• We-Are-Coming Post • Sponsor post + Progress Post • Trade Display Coreflute colour design
Riley	• Trade Display Design Finalised • Desk Frame Design Done • Amount of material required details finalised - Coreflute and metal required final	• Trade Display Development Brief • Trade Display desk shelf design
Liam	• Final wheel design finalised • Final Bearings decided and ordered	• Cut Final Car • Get Paint and Primer • Fully Finalise the Car - NEARLY DONE
	EXPENDITURE	INCOME & Donations
Finance - Income & Expenditure	Accomodation - 1478.78	\$280 - Rajpal Family
	Fees Paid - \$1600.00	1600 - Trojan
	Uniforms - \$280	1000 - Lions Hellenic, NEMR
	Bearings - 120.38	2200 - MHS, Mrs. Costa (Principal) donation
		672 - MHS, Casual Day
Total (Spent, Income)	\$ 3,478.78	\$ 5,952.00
Total Remaining	\$	2,473.22
	Conclusion	
Conclusion	February remained a productive month, most of Enterprise and Engineering are done, Compliance is about to initiate once car design is Fully finalised, Funds raised, major payments made. Trade Display design finalised and frame nearly built, coreflute design nearly done. Uniforms ordered, beanies here. Overall, productive month.	

Figure 39 - Risk Matrix Table - Excel - Credit Vivaan.R Velocity Vortex

Risk Management Table

Identified Risk	Probability	Severity	Risk Rating	Preventative Control	Mitigation	Impact on
Technical Performance Risk: Car fails to meet predicted performance before racing	Possible	MAJOR	EXTREME	Continuous CFD evaluation, prototype testing and mentor review	Make quick design adjustments and optimise setup before racing	Resources: Higher Number of resources - increased fund requirement. Timing: Reduced Testing time. Scope: May limit ability to implement advanced design features Quality: Lower race performance and Engineering score
Breach of Competition regulations	Unlikely	MAJOR	EXTREME	Competition regulation check list and internal scrutineering checks	Thorough proofreading/checking then making Immediate corrections before submission	Resources: Re-manufacturing or Re-printing of items - higher costs Timing: Delays before final submission Scope: Certain features may need removal to comply Quality: Loss of marks and penalties.
Manufacturing or supply delays	Possible	MODERATE	MODERATE	Early ordering and timeline modifications	Adjust schedule and prioritize components	Resources: Lower resource availability for deliverables Timing: Shorter production and testing phase Scope: Reduced ability to quality control manufacture Quality: Reduced finishing quality.
Academic work load effecting deadlines	Highly Possible	MODERATE	MODERATE	Align Gantt chart with school Calendar	Redistribute work load and extend work sessions	Timing: Shorten preparation window Scope: Optional improvements may not be possible Quality: Less polished Trade Display or Portfolios.
Insufficient Sponsorship funding	Possible	MODERATE	MODERATE	Initiating sponsor acquirement period earlier, responsible budgeting	Re-allocate budget to prioritize car performance, seek assistance from school through fundraisers	Resources: Limited for specific deliverables Timing: Possible delays in purchases Scope: Reduced scope of trade Display or merchandise Quality: Lower deliverable or material quality
Car Damage	Possible	MAJOR	EXTREME	Pre submission inspection and careful transport, bring additional back-up parts	Use spare parts and rapid repairs	Resources: Cost of replacement components Timing: Reduced car racing times Scope: Will influence out of scope factors Quality: Reduced race performance
Miscommunication	Unlikely	MAJOR	EXTREME	Ongoing Communication regarding assigned work and deadlines	Reassign tasks and clarify all doubts	Resources: No influence Timing: Increased time to produce deliverables Scope: Reduce scope of specific deliverables Quality: Lowered deliverable quality

Figure 40 - Risk Management Table - Excel - Credit Vivaan.R Velocity Vortex

Risk Matrix

Risk Rating	Associated Action if the risk occurs
EXTREME/HIGH	Immediately contact team leader and supervising teacher who will provide utmost priority to the risk and initiate given mitigation for the risk.
MODERATE	Contact team manager who will assess the risk and provide priority based on the threat level, and subsequently proceed with the given mitigation.
Low	Contact Team leader who will review the threat over the course of the week and undergo the mitigation through that week (Or subsequent week based on threat level)

Figure 41 - Risk Matrix Table - Excel - Credit Vivaan.R Velocity Vortex

Monitoring, Controlling & Scope Creep

Velocity Vortex maintained structured weekly Status Reports aligned with our Gantt chart, recording each team member's completed tasks and pending deliverables. The monthly reports also allowed for detailed financial tracking, running parallel to our budget as they tracked each month's cash inflow and outflow. Once tasks reached closure they required evidence-based sign-off, ensuring accountability and quality control. However, **Scope Creep** was identified as our schooling schedules showed signs of impacting the project, and the Trade Display and final car renders exceeded initial time allocations due to multiple refinements. In response, high-priority deliverables were protected, responsibilities redistributed, and internal deadlines adjusted. This kept the team on track and ensured a polished outcome.



Mentorship

Industry mentorship was an integral part of our STEM Racing journey. Through regular engagement, our mentors’ support, industry knowledge and expertise guided us while strengthening our technical, professional, and collaborative skill sets. The list below outlines our mentors, the team members they supported and the key skills developed through their guidance.

Vivaan Rajpal - Team Manager

Mentors:

Don Grant -BHP - Project Management and Risk Assessment
Stephen Read: STEM Racing Educator & Mentor - Engineering & CAD
Kahlan Mashford - UniSA HR Management, Alumni - Acrux Team Manager
Deepak Rajpal - Accountant/Finance Manager
Mary-anne Richards - Sponsor acquirement

Skills Developed:

Project Management
Leadership and Effective Communication
Sponsor Acquisition techniques, Engagement and Management
Detailed Finance management, Account handling and tracking
Detailed Orthographic Drawing development

Liam Bairstow - Lead Engineer

Mentors:

Stephen Read - SA STEM Racing rep - Engineering & CAD
Mason Green - Fitter & Turner AusRAMD , Alumni -ACRUX Engineer
Karl Read - STEM Racing teacher
David Chan - Adelaide University

Knowledge Gained

Car manufacture and testing
Aerodynamic principles
Technical skills in Engineering Software (Fusion 360, CFD, VWT)
Problem solving and Effective Communication
Manufacturing/3D Printing

Sebastian Richards - Graphic Designer

Mentors:

James Coulter - Graphic designer -JC Designs
Jack May - Dynamic Racing Social Media and Finance Manager
Karl Read - STEM Racing teacher

Knowledge Gained:

Brand Application and Design Software proficiency
Uniform Design, Logo creation and Colour Theory
Typography - art and techniques
Social Media Management and customer engagement

Riley Hutchinson - Trade Display Manager

Mentors:

Mason Green - Fitter & Turner AusRAMD , Alumni -ACRUX Engineer
Tahnee Blight - Adelaide University Marketing and Branding
Karl Read - STEM Racing teacher
Scott Moritz - Senturion Steel

Knowledge Gained:

Material Science and Impact of Choices
CAD Proficiency
Creativity and Innovation in Design Processes
Brand Application

Overview

Our mentors elevated our project through specialised knowledge, industry insight, and practical guidance, strengthening our technical and enterprise skills while enhancing the professionalism of our work.

Linking Skills with Future Careers

Participation in STEM Racing provided each team member with opportunities to develop both industry-specific and employability skills. Through hands-on roles in engineering, leadership, design and project management team members strengthened technical capabilities, professional competencies, and problem-solving skills. These skills are directly applicable to their chosen career pathways, preparing them for future roles in fields such as medicine, engineering, aviation and software development, while also fostering adaptability and lifelong learning in professional contexts.

Vivaan - Team Manager

Future Career: General Practitioner

Industry-Specific Skills

- Decision-making under pressure
- Analytical thinking
- Interpreting complex formation
- Risk assessment and problem identification

Employability Skills

- Leadership and team coordination
- Professional communication
- Conflict resolution
- Time and workload management and organisation

Evidence:

Co-ordinated team meetings, delegated tasks, resolved challenges, and managed project timelines.

Career Link

These skills align with the role of a General Practitioner, where professionals communicate effectively with patients, analyse information, assess risk and make informed decisions within collaborative healthcare teams. They also align with Logistic Officer pathways in the ADF, supported by a Bachelor of Project Management at Adelaide University.

Liam - Lead Engineer

Future Career: Mechanical Engineer

Industry-Specific Skills

- CAD modelling and design optimisation
- Engineering analysis and performance testing
- Iterative design development
- Date-driven technical problem solving

Employability Skills

- Critical thinking
- Collaboration with multidisciplinary teams
- Technical communication
- Project planning and organisation

Evidence:

Developed and tested car components using CAD, FEA and CFD, refining designs based on performance data.

Career Link

These skills align directly applicable to Mechanical Engineering, where engineers design, analyse and refine systems to improve efficiency and performance, and also support Mechanical Engineer pathways in the ADF via a Bachelor of Engineering (Mechanical) at Adelaide University.

Sebastian - Graphic Designer

Future Career: Aviation/Pilot

Industry-Specific Skills

- Visual information interpretation
- Spatial awareness
- Interpreting complex visual information
- Precision and attention to detail
- Situational awareness

Employability Skills

- Communication through visual media
- Creative thinking and innovation
- Adaptability
- Problem solving

Evidence:

Produced portfolio graphics, branding assets, and marketing visuals requiring precision, consistency and complex information interpretation.

Career Link

These abilities are highly relevant in aviation, where pilots interpret complex instrument data, maintain situational awareness, visual processing, and precision, and also support Strategic Communications and Media Officer pathways in the ADF via a Communications and Media degree at Flinders University.

Riley - Trade Display Manager

Future Career: Software Engineering/Programming

Industry-Specific Skills

- Systems thinking
- Technical design and integration of digital elements
- User interface awareness
- Logical problem solving

Employability Skills

- Creative problem solving
- Collaboration
- Project Planning
- User-focused design thinking
- Organisation and logistics management

Evidence:

Designed and integrated digital and physical elements of the trade display, managing workflow and user interaction.

Career Link

These skills align with software engineering careers where programmers require logical thinking, design digital systems, solve technical problems and create effective user focussed experiences, and also support Cyber Systems Specialist or System Technician pathways in the ADF via an IT degree AIE or Adelaide University.

Overview

Through STEM Racing, the team developed a combination of industry-specific and employability skills that will support their transition into future careers and prepare them for real-world professional environments, while fostering lifelong learning and adaptability.

WEBSITE REFERENCES FROM EACH SLIDE

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